

First-birth housing: retain the target after the check

Calibration assessment for the September 14 presentation

Prepared for Tommaso De Santo

5 September 2026

Decision after the completed check

Retain the existing 0.720246-room target and return attention to the model. The actual Sun-Abraham interacted regression reproduces the target to machine precision. Changing only the reference choice for the 1986 first-birth cohort gives **0.730459 rooms**, with the same 49,457 observations, 4,112 clusters and fitted housing values equal to within **4.41e-12 rooms**. The normalization concern is real; its effect in this test is **+0.010213 rooms**. This is not an estimate of total bias or a new target.

The predeclared descriptive comparisons use the same 27 birth cohorts and the same weights at both endpoints. They give **0.811582**, **0.797759** and **0.804369 rooms**, respectively, using pre-endpoint, post-endpoint and averaged shares. Each is unchanged by the tested reference choice to machine precision. They have no newly estimated standard error and are not replacements for the approved target. They support a substantial housing response rather than a measurement explanation of the model shortfall.

The earlier suggestion that this issue might account for the model's miss was premature. The current model, twelve targets, weights and production calibration remain unchanged. The empirical aggregation convention should be documented for the paper; these checks do not certify causal interpretation. The appendix retains every target-fit row and parameter for production and the verified refinement candidate. The practical next step is to assess the candidate and compare its policy implications under the unchanged target system.

Why the reference period matters

A birth cohort is the group of women with a first birth in the same calendar year. The estimator fits a separate housing profile for each cohort and then averages those profiles using the cohort shares observed at each event time. The reference period, two years before first birth, is assigned a zero effect.

The sample-only replay finds **16 birth cohorts with no observation at event year -2**. Such cohorts account for **41.91%** of the cohort-share weight at -1 and **46.01%** at +3. A second sample-only check verifies that all regression inputs are nonmissing, first-birth cohort is constant within person, and the included event indicators cover every treated observation except -2. These checks require no regression coefficients.

For a cohort without the reference observation, shifting all its observed housing coefficients up by one room and its person fixed effects down by one room leaves every fitted observation unchanged. If that cohort receives different weights at the two endpoints, the weighted difference changes. That is a failure of invariance to an arbitrary normalization, beyond the usual question of whether changing cohort composition is economically desirable.

A concrete check of the normalization

Let $\delta_{g,k}$ denote the cohort- g housing coefficient at event time k , and $w_{g,k}$ its cohort share at that event time. The reported target is

$$D = \sum_g w_{g,3} \delta_{g,3} - \sum_g w_{g,-1} \delta_{g,-1}.$$

For a cohort with no -2 observation, the transformation $\delta_{g,k} \mapsto \delta_{g,k} + c_g$ on its supported event cells and $\alpha_i \mapsto \alpha_i - c_g$ for its members' person fixed effects leaves the regression's predictions unchanged. The reported contrast, however, changes by

$$\Delta D = (w_{g,3} - w_{g,-1})c_g.$$

The **1986 first-birth cohort** supplies a numerical example with observations at both endpoints. It has no -2 observation; its weights are **0.026299** at -1 and **0.040435** at +3. A one-room normalization shift changes the contrast by **0.014136 rooms**, while the verified change in every fitted row is **zero**. The one-room shift is an algebraic illustration, not an economic intervention, an estimate of bias, or a proposed correction.

The installed estimator calculates these cohort shares before the regression's singleton pruning. The all-inputs-observed assertion verifies that saved input support is the share-calculation sample. The paired Torch regression now verifies the actual coefficient change and identical final estimation membership, beyond the earlier design identity. The cohort-specific coefficients and final support are retained in the follow-up packet. The primary receipt has 49,457 estimation rows; its input sample contains 49,872 rows.

A common set of cohort weights applied to estimable within-cohort changes would cancel this particular arbitrary level. That is a candidate diagnostic repair, not an approved replacement target: cohort overlap, remaining rank conditions, uncertainty and economic interpretation must all be checked. No target is dropped, demoted or reweighted in this review.

What passed, and what did not

Source and target provenance verification passed, as did two sample checks. The two-regression loop passed on synthetic data locally and on Torch. Full Torch job **17007732** completed in **8m03s**, reproducing the primary point and verifying fitted-value and sample invariance under the changed reference.

The earlier 900-second full-covariance replay and a 1,200-second local paired-fit attempt timed out. The successful Torch run supersedes those computational blockers. It skips the costly cohort-share covariance calculation, which is not needed for the point-estimate invariance test; no new target standard error is claimed. It uses Stata 19 in version-17 mode, pinned installed ado sources and a hash-matched minimal analysis sample. The primary builder and target outputs are unchanged. Only aggregate results are retained; temporary person-level working copies used on the author's cluster are removed after collection.

The event-time pattern does not supply a convenient replacement

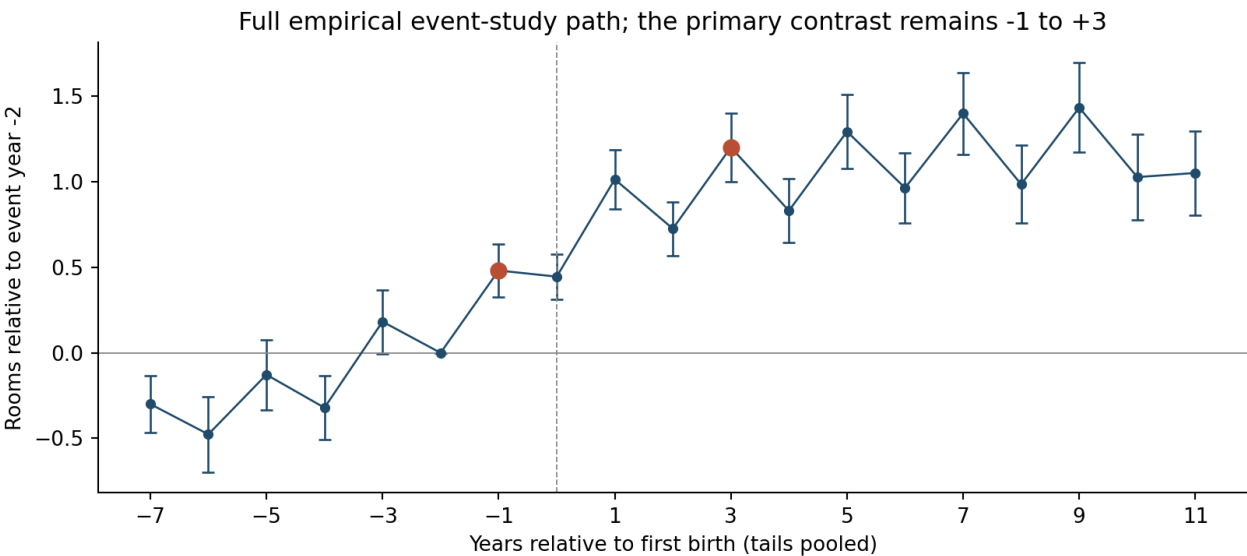


Figure 1: Supplemental empirical profile from the unchanged production event table. Bars show its existing pointwise 95 percent intervals, conditional on the reported specification and normalization.

The profile alternates substantially between odd and even event years. However, nearby four-year summaries of the saved coefficients remain near 0.72 rooms:

Contrast	Rooms	Interpretation
-1 to +3	0.720246	Existing target; reported SE 0.085260
-2 to +2	0.726160	Diagnostic; reported SE 0.080315
Mean of +2/ +3 minus mean of -2/ -1	0.723203	Diagnostic; joint uncertainty unavailable
0 to +4	0.385433	Different object, excluding adjustment through the birth year

These comparisons are arithmetic diagnostics from the old coefficient table, not newly estimated or validated causal effects. No new joint uncertainty calculation for these alternative windows was made. In particular, the smaller 0-to-4 value is not a reason to replace the target with a number closer to the model.

The treated input observations span **1970-2019**. The weighted mean first-birth year is **1994.64** at -1 and **1992.97** at +3; the share of weight on birth cohorts from 2007 onward falls from **21.39% to 18.05%**. The total variation distance between endpoint cohort shares is **0.20236**: 20.24 percentage points of cohort probability weight would have to be reassigned to make the two mixes identical. These are cohort-share input statistics, not claims about the final regression sample or a quantified correction to the target.

What the model measures

The active housing statistic compares two copies of the households that successfully have a first birth. One copy has the baby; the other remains childless. Both begin with the same wealth, age, income, tenure and location, and follow their respective saving and housing choices over one four-year model period. The difference in their destination rooms is also a difference in changes because their starting states are identical.

At the review candidate, the 2019-2023 treated copy occupies **5.800342 rooms** and the childless copy **5.334298**, giving **0.466044 rooms**. The treated branch already permits a later birth at the destination date: continuation births equal **26.88%** of its mass. Therefore, a missing second-birth option in this active measurement is not the explanation for the shortfall. An older same-policy helper has different timing and must not be confused with this dated calculation.

Comparison	Empirical target	Model statistic
Population	PSID women who are reference persons or spouses, with one observation per household-year	Successful first-birth households in the fitted first-birth risk set
Control	Women confirmed childless in their observed histories	Identical copies held childless through the window
Clock	First-birth event years -1 and +3	One four-year interval, 2019-2023 for the fitted terminal moment
Calendar coverage	Pooled historical PSID interviews	The fitted terminal four-year cohort
Before-birth differences	Estimated from an observational panel; anticipation and selection remain relevant	Starting-state differences between copies are zero by construction
Later births	Allowed in treated mothers' histories	Allowed in the treated branch at the destination

Matching the duration does not establish that these are identical economic objects. In particular, the model comparison holds the initial state fixed for households selected into first birth, whereas the empirical estimator uses a separate childless population. Neither exact reproduction nor a small standard error establishes the missing comparability assumptions. No new estimate of the size of this model-to-data selection discrepancy is available from this replay.

What this means for calibration

The best review candidate reduces the unchanged twelve-target loss from **30.482967** to **23.153445**. Across the **35 coordinate and joint evaluations**, the largest first-birth response is **0.466044 rooms** and the smallest average housing level is **6.270186 rooms**. Those extrema do not occur at the same parameter vector. The target pair is **0.720246** and **5.779970 rooms**. This neighborhood contains a meaningful improvement but provides no evidence that the two housing discrepancies have been jointly resolved. It is not a proof that the target pair is globally unreachable.

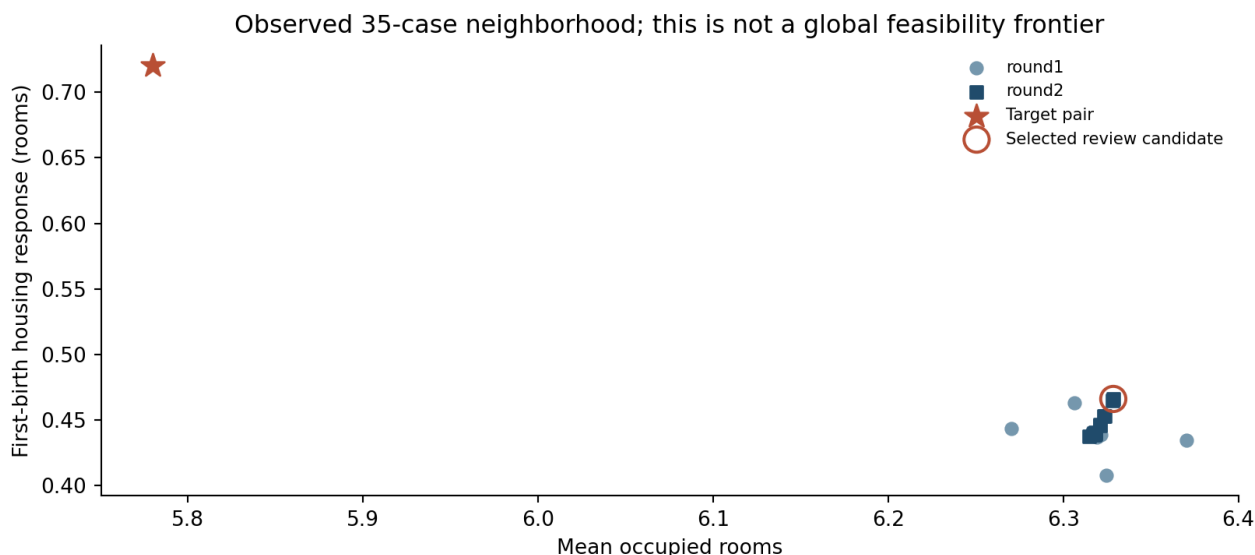


Figure 2: Supplemental housing-fit comparison using saved evaluations. No new model solution is computed.

The remaining model measurement comparison should reproduce a stated empirical population and event clock in the model, including before-birth housing choices, and compare that statistic with the current matched-copy response. It should preserve the full twelve-moment fit table and distinguish a measurement change from an improvement under the existing objective. A new empirical weighting rule requires its own uncertainty calculation and target-contract decision.

Retain both the production calibration and the verified refinement candidate for the next fit-and-policy comparison. The principal question is how much of the rooms shortfall reflects housing-demand parameters, and how much reflects comparing different populations and timing conventions. Another broad search alone cannot answer it. Existing policy numbers belong to the retained production calibration; this report supplies no revised policy or welfare result.

Complete target fit: verified candidate

Gaps equal model minus target. Contributions are weight times squared gap. The weights retain their existing empirical or synthetic interpretations; this loss is not a specification-test statistic.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.921698	+0.003698	1425.74	0.0195
Childlessness	0.188000	0.189859	+0.001859	17180.7	0.0594
Mean age at first birth	26.044627	26.251703	+0.207076	44.4444	1.9058
First births at age 30+	0.260327	0.237163	-0.023165	10000	5.3660
Rooms response to first birth	0.720246	0.466044	-0.254202	137.565	8.8893
Rooms, 3+ minus 1-2 children	0.367700	0.379198	+0.011499	2958.51	0.3912
Family ownership gap	0.167662	0.169578	+0.001917	14229.6	0.0523
Ownership, ages 30-55	0.575472	0.546618	-0.028854	1207.85	1.0056
Mean occupied rooms	5.779970	6.328594	+0.548624	11.9732	3.6038
Wealth / annual earnings	6.873100	7.031196	+0.158096	6.28767	0.1572
Bequest flow / wealth	0.008800	0.008485	-0.000315	5.16529e+06	0.5139
Older wealth/income p90 / p50	3.448111	3.303595	-0.144516	56.9598	1.1896

Complete target fit: retained production

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918000	1.922907	+0.004907	1425.74	0.0343
Childlessness	0.188000	0.189407	+0.001407	17180.7	0.0340
Mean age at first birth	26.044627	26.256032	+0.211404	44.4444	1.9863
First births at age 30+	0.260327	0.237579	-0.022748	10000	5.1748
Rooms response to first birth	0.720246	0.436418	-0.283828	137.565	11.0821
Rooms, 3+ minus 1-2 children	0.367700	0.403441	+0.035741	2958.51	3.7793
Family ownership gap	0.167662	0.161699	-0.005963	14229.6	0.5060
Ownership, ages 30-55	0.575472	0.544488	-0.030984	1207.85	1.1596
Mean occupied rooms	5.779970	6.317291	+0.537321	11.9732	3.4568
Wealth / annual earnings	6.873100	6.932652	+0.059552	6.28767	0.0223
Bequest flow / wealth	0.008800	0.008433	-0.000367	5.16529e+06	0.6971
Older wealth/income p90 / p50	3.448111	3.236511	-0.211600	56.9598	2.5503

Complete parameter comparison

Parameter	Production	New candidate	Bounds / restriction	Near bound?
Annual patience β	0.995117	0.995739	[0.94, 0.9995]	No
First-birth dispersion κ_1	2.168173	2.168173	[0.02, 50]	No
Later-birth dispersion κ_C	1.736471	1.736471	[0.02, 50]	No
Owner-service premium χ	1.043472	1.043472	[0.1, 5]	No
Supply intercept H_0	14.562959	14.562959	[0.2, 80]	No
Bequest strength θ_0	0.528428	0.549189	[0, 8]	No
Bequest shifter θ_1	0.107249	0.107249	[0.02, 16]	Raw scale only
Per-child room floor	0.282210	0.248877	[0.1, 1.8]	No
First-birth fixed cost	4.559138	4.559138	[0, 8]	No
First-child room intercept	0.364931	0.464931	[0, 0.5]	No
2007-2023 child-value change	-0.328714	-0.328714	[-1.5, 0.2]	No
Tenure dispersion κ	0.005000	0.005000	Externally fixed	Not estimated
Dated supply elasticity η	0.630000	0.630000	Externally fixed	Not estimated

The old fertility-preference intercept is re-normalized to replacement for every parameter vector. New candidate values are $\psi_{2007} = 0.28947879$ and $\psi_{2023} = -0.03923512$. The stored near-bound flag uses raw parameter units. For the bequest shifter, interpret it alongside its position in the logarithmic search interval; a raw-scale flag alone does not establish a binding search constraint.

The supply restriction 0.63 applies to dated markets. The inherited initialization retains elasticity 1.75 before the dated supply intercept is normalized. Both values and that order are unchanged across this comparison; whether this is the intended economic contract remains a discussion item in the separate code audit.

Evidence and reproduction

All paths below are relative to the project root: `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26`.

- Primary estimator: `code/data/psid_followup_mar2026/sa_rooms_first_birth_household_aligned_v1.do`, sample construction and event definitions at lines 65-147, estimator and contrast at lines 166-212.
- Primary target and original event table: `code/data/psid_followup_mar2026/output/sa_rooms_first_birth_household_aligned_v1/`.
- Sample replay driver: `code/data/psid_followup_mar2026/audit_first_birth_rooms_measurement.py`.
- Aggregate analysis and two supplemental figures: `code/model/tools/analyze_e5f_first_birth_measurement.py`.
- Complete receipts, aggregates, null-direction check, initial failed local receipts, completed paired Torch check and independent review: `output/model/e5f_first_birth_measurement_review_20260905a/`.
- Active matched-branch implementation: `code/model/tools/run_e5f_transition_calibration.py`, functions `begin_dated_first_birth_housing_branch` and `finish_dated_first_birth_housing_branch`.
- Saved candidate: `output/model/e5f_morning_refinement_20260905a/round2/task_009/`; its dated-branch CSV is under `cases/task_001/` within that folder. This inner case identifier does not change the selected outer candidate.
- Earlier complete calibration readout and standard seventeen-graph packet: `docs/model/e5f_bounded_calibration_refinement_review.md` and `output/pdf/e5f_bounded_calibration_refinement_review.pdf`.

The installed estimator was also inspected at `/Users/tommasodesanto/Library/ApplicationSupport/Stata/ado/plus/e/eventstudyinteract.ado`: cohort-share estimation at lines 40-47, interacted regression at line 85, and weighted aggregation at lines 103-108. The evidence manifest pins its hash.

Regenerate the aggregate analysis without a Stata regression or model solve:

```
python3 code/model/tools/analyze_e5f_first_birth_measurement.py
```

The output-folder README retains the original regression budget, both successful sample checks, the initial failed local attempts, successful Torch follow-up and PDF build command. Full per-case target and parameter tables are retained alongside this report. The two new figures are supplemental; the standard model diagnostic packet is unchanged.